

Toran Bahadur Niroula

Aerospace Engineer

Lalitpur, Nepal | +977 9742302381 | 078bas043.toran@pcampus.edu.np
toranniroula.com.np | linkedin.com/in/toranbdrniroula | github.com/Toran999

Research Profile

Aerospace engineer specializing in experimental and computational fluid mechanics. Experience spans quantitative Rainbow Schlieren Deflectometry, PIV/POD flow analysis, compressible-flow CFD in OpenFOAM, and development of open-source Python research tools. Research interests: high-speed and hypersonic flows, turbulence, experimental fluid mechanics, and data-driven methods for CFD.

Education

Bachelor of Engineering in Aerospace Engineering

2022–2026

Institute of Engineering, Pulchowk Campus, Tribhuvan University

Lalitpur, Nepal

- **Aggregate:** 82.62% (Distinction); ranked 4th in the graduating class
- **Coursework:** Fluid Dynamics, Compressible Aerodynamics, Hypersonic Aerodynamics, Advanced Propulsion, Computational Fluid Dynamics, Numerical Methods, Aircraft Structures, Flight Dynamics, Control Systems

Research Experience

Research Collaborator

2026–present

BioFSI Lab, University of Birmingham

Remote

- Investigating flight mechanics and vortex formation of **dandelion seeds** using CFD with overset-mesh methods.
- Developing fluid–structure interaction approaches to model the flexible pappus and its influence on aerodynamic behavior.
- Developing an OpenFOAM study of **unsteady wing–propeller aerodynamic interaction**, focusing on wake development and aerodynamic coupling, under group mentorship.

Research Intern, Experimental Aerodynamics Group

Oct–Nov 2025

Indian Institute of Technology Kanpur

Kanpur, India

- Analyzed **500 PIV snapshots** of a NACA0015 airfoil in reverse flow at 12° angle of attack ($Re = 1.35 \times 10^5$) using PIVlab and Python/MATLAB workflows.
- Extracted mean velocity, vorticity, Q-criterion, turbulence intensity, and Reynolds-stress fields.
- Applied **Proper Orthogonal Decomposition (POD)** to identify dominant coherent structures in the separated wake.
- Gained hands-on experience with tuft visualization and aerodynamic force/torque measurements in rotorcraft flow-control studies.

CFD Research Intern, FOSSEE Fellowship

Feb–Jun 2025

Indian Institute of Technology Bombay

Remote

- Conducted an OpenFOAM case study (**simpleFoam**, Spalart–Allmaras, **snappyHexMesh**) of a fixed-wing UAV with blended winglets.
- Performed mesh-independence and turbulence-model sensitivity studies; compared reproduced baseline results against the reference study.
- Extended the study to multiple winglet cant angles, assessing effects on lift-to-drag ratio and wingtip vortex structure.

Research Projects & Software

Rainbow Schlieren Deflectometry of a Heated Compressible Wall Jet 2025–2026

Undergraduate Thesis, Pulchowk Campus

- Designed and built a **Rainbow Schlieren Deflectometry (RSD)** facility to measure density-gradient fields around a heated compressible wall jet over a temperature-controlled flat plate.
- Developed image-processing and reconstruction methods to convert raw RSD images into optical displacement, density-gradient, and density fields.
- Compared experimental results with CFD and used a natural-convection case as an independent validation dataset.

pyrsd: Python Package for Rainbow Schlieren Deflectometry Data 2026

Open-source research software

doi:10.5281/zenodo.19219233

- Developed a Python package for processing and quantitative analysis of RSD data; archived on Zenodo.

Shock Interactions and Scramjet Intake 2026

Independent CFD study

- Studied shock–shock interactions with `rhoCentralFoam`, comparing numerical pressure distributions against local-inclination and shock-polar predictions.
- Designed and evaluated a single-ramp scramjet intake at Mach 6 using inviscid and RANS simulations.

Engineering Experience

NIMBUS: Autonomous Agricultural UAV 2026

Preliminary design and flight dynamics

- Contributed to conceptual sizing and configuration of an electric agricultural UAV for a 200 kg payload; performed aerodynamic, performance, and flight-dynamics analysis.

Twin-Boom Fixed-Wing UAV 2025

Design and fabrication, three-member team

- Performed XFLR5 aerodynamic analysis, hot-wire foam cutting, and flight testing.

Design Team Member, DBF Pulchowk 2023–2024

AIAA Design/Build/Fly

IOE Pulchowk

- Contributed to aerodynamic and stability design of the competition aircraft *Junkiri*: sizing, stability analysis, flight testing, and technical documentation.

Technical Skills

CFD & Simulation	OpenFOAM (rhoCentralFoam, simpleFoam, snappyHexMesh), ANSYS Fluent, ParaView
Experimental	Rainbow Schlieren Deflectometry, Schlieren, Particle Image Velocimetry, tuft visualization
Flow Analysis	PIVlab, POD, modal analysis, vorticity, Q-criterion, Reynolds-stress analysis
Programming	Python (NumPy, Matplotlib), MATLAB, L ^A T _E X
CAD & Design	CATIA, SolidWorks, XFLR5, X-Plane

Teaching Experience

Aerodynamics Practical Instructor 2025–2026

Department of Aerospace Engineering, IOE Pulchowk Campus

Lalitpur, Nepal

- Conducted laboratory sessions in aerodynamics for third-year students.

Guest Lecturer, Instrumentation Systems 2025–2026

Department of Aerospace Engineering, IOE Pulchowk Campus

Lalitpur, Nepal

- Delivered guest lectures on instrumentation systems to second- and fourth-year students.